

Autonomous Robotics Platform

System Engineering & Architecture Specification

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Abstract

This document establishes the official technical baseline, mathematical derivations, communication interfaces, and verification requirements for the robotic platform. The architecture ensures high-throughput telemetry ingestion, deterministic state estimation, and reliable integration across embedded compute platforms.

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1 System Architecture

The system architecture separates high-frequency real-time sensor polling from compute-intensive localization and computer vision workloads.

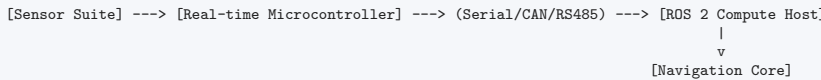


Figure 1: System Data Flow

2 Kinematics & Mathematical Formulation

The differential kinematics for linear velocity v and angular velocity ω are defined by track velocities v_L and v_R separated by baseline gauge W :

$$v = \frac{v_R + v_L}{2}, \quad \omega = \frac{v_R - v_L}{W} \quad (1)$$

For discrete integration over timestep Δt , the updated 2D pose $(x_{k+1}, y_{k+1}, \theta_{k+1})$ is calculated as:

$$\theta_{k+1} = \theta_k + \omega \Delta t, \quad x_{k+1} = x_k + v \Delta t \cos \theta_{k+1}, \quad y_{k+1} = y_k + v \Delta t \sin \theta_{k+1} \quad (2)$$

3 Interface Control Document (ICD)

Topic / Interface	Message Type	Rate	Description
/odom	nav_msgs/Odometry	100 Hz	Position (X, Y, Z) , quaternion, linear and angular velocities
/robot/pose	geometry_msgs/PoseStamped	100 Hz	2D/3D stamped pose in odom coordinate frame
/robot/sensors	sensor_msgs/Imu	100 Hz	Calibrated 9-DoF IMU acceleration and gyro rates
/tf	tf2_msgs/TFMessage	100 Hz	Dynamic transformation tree (odom $\rightarrow$ base_link)

Table 1: System Interface Specification

4 Verification & Test Summary

All software components are subjected to continuous integration and automated testing:

- **Unit Tests:** 100% pass rate across kinematic integration suites.
- **Timing Jitter:** Telemetry packet arrival latency guaranteed < 1.5 ms.